

Chapter 1 — Introduction to Python

Original educational notes prepared as sample source material for the RAG assignment.

What is Python?

Python is a high-level, general-purpose programming language known for readable syntax and a large standard library. Python programs are commonly used for automation, web development, data analysis, artificial intelligence, scientific computing, and education.

Python Syntax

Python uses indentation to organize blocks of code. A statement can be as simple as `print("Hello, Python!")`. Python is dynamically typed, which means a variable does not need a type declaration before a value is assigned to it.

Comments

Comments help explain code to a reader. A single-line comment begins with the `#` character. Comments are ignored by the Python interpreter.

Basic Program

A simple Python program can assign a value to a variable and display it. For example: `name = "Rameesha"` followed by `print(name)`. The `print` function writes information to the output.